

MULTI-ASSET PORTFOLIO OPTIMIZATION USING HYBRID CLASSICAL AND QUANTUM COMPUTING

WISER Vanguard Quantum Challenge 2026

Developed by

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Tech Stack

Python | Qiskit | QAOA | SciPy | CPLEX | Streamlit | Plotly | Gemini AI

Year

2026

Abstract

Portfolio optimization is one of the most important applications of mathematical optimization in quantitative finance. Traditional optimization techniques are capable of generating efficient investment allocations but often become computationally expensive when additional real-world investment constraints such as transaction costs, liquidity requirements, diversification, and sector allocation limits are incorporated. As the number of financial assets increases, solving these constrained optimization problems becomes significantly more challenging.

This project presents a Hybrid Classical–Quantum Multi-Asset Portfolio Optimization framework that integrates modern portfolio theory with Quantum Approximate Optimization Algorithm (QAOA) to address complex portfolio selection problems. Historical financial market data is collected and processed to estimate expected annual returns, covariance matrices, liquidity scores, transaction costs, and asset correlations. These financial metrics are transformed into a Quadratic Unconstrained Binary Optimization (QUBO) model, enabling the portfolio selection problem to be represented as an Ising Hamiltonian suitable for execution using quantum optimization techniques.

The proposed solution employs a two-stage optimization strategy. In the first stage, QAOA identifies an optimal subset of assets by solving the binary optimization problem. In the second stage, a classical continuous optimization algorithm refines the investment weights allocated to the selected assets while satisfying practical portfolio constraints. This hybrid methodology combines the global search capability of quantum optimization with the precision of classical optimization.

To improve interpretability and usability, an interactive Streamlit dashboard has been developed. The dashboard provides real-time visualization of portfolio allocations, expected returns, portfolio risk, transaction costs, liquidity analysis, diversification metrics, QAOA performance, objective function decomposition, and classical versus quantum optimization comparisons. An AI-powered Portfolio Copilot has also been integrated using Gemini AI to generate intelligent portfolio insights, answer user queries, and automatically produce comprehensive PDF reports summarizing optimization outcomes.

The project demonstrates how hybrid quantum-classical optimization techniques can be applied to real-world financial portfolio management problems while providing an intuitive visualization platform for investors, researchers, and practitioners. The developed framework serves as a scalable foundation for future quantum financial applications and highlights the practical potential of quantum optimization in investment decision-making.

Introduction

Portfolio optimization seeks to maximize returns while minimizing risk under real-world constraints like transaction costs, liquidity, and sector limits. Markowitz's mean-variance framework laid the foundation for this, but as constraints multiply, finding globally optimal solutions becomes computationally hard.

Quantum computing offers a new approach to these combinatorial problems. The Quantum Approximate Optimization Algorithm (QAOA) is well-suited to Quadratic Unconstrained Binary Optimization (QUBO) problems, and portfolio selection naturally maps onto this formulation.

This project builds a **Hybrid Multi-Asset Portfolio Optimization framework** that pairs quantum and classical methods: QAOA selects an optimal asset subset under QUBO constraints (cardinality, sector exposure, liquidity, turnover), while a classical optimizer (SLSQP) determines precise capital allocation across the selected assets.

An interactive **Streamlit dashboard** visualizes allocations, optimization metrics, and classical-vs-quantum comparisons, supported by an AI-powered Portfolio Copilot that generates explanations, recommendations, and downloadable PDF reports.

The goal is to demonstrate quantum optimization's applicability to portfolio management while delivering a practical, explainable decision-support tool.

Problem Statement

Source : WISER Vanguard Quantum Challenge 2026

Multi-Asset Portfolio Construction

- Given return assumptions, risk estimates, cost estimates, and portfolio constraints, recommend an allocation that improves expected utility while staying inside predefined investment guardrails.
- An example optimization workflow is:
- Inputs (Forecast + Investment Goals) → Optimization Solver (Constraints + Objective Function) → Outputs (Optimal Asset Allocation + Investment Rationale)

Problem Description

Investment firms continuously seek to construct portfolios that maximize expected returns while maintaining acceptable levels of risk. In practice, portfolio construction involves balancing several competing objectives, including return maximization, risk minimization, transaction costs, liquidity requirements, diversification, and sector allocation limits.

Traditional portfolio optimization techniques become increasingly complex as the number of financial assets and investment constraints grows. Portfolio managers must evaluate numerous possible asset combinations while ensuring that all investment policies and regulatory constraints are satisfied. This results in a computationally intensive optimization problem that is difficult to solve efficiently for large portfolios.

The WISER Vanguard Quantum Challenge addresses this real-world problem by encouraging the development of advanced optimization techniques capable of producing high-quality portfolio allocations while satisfying practical investment constraints. The challenge emphasizes not only identifying profitable investment opportunities but also ensuring that the resulting portfolio remains within predefined financial guardrails.

Project Objectives

The primary objective of this project is to develop a Hybrid Classical-Quantum Multi-Asset Portfolio Optimization Framework that generates optimal investment allocations while satisfying practical financial constraints. The project aims to combine classical optimization techniques with quantum computing to improve portfolio decision-making and demonstrate the applicability of quantum optimization in financial engineering.

The specific objectives of the project are:

1. To collect and preprocess historical financial market data for multiple assets.
2. To calculate key financial metrics, including expected returns, covariance matrices, asset correlations, liquidity scores, and transaction cost estimates.
3. To design a custom portfolio objective function that simultaneously considers expected return, portfolio risk, transaction costs, liquidity, diversification, and cash utilization.
4. To formulate the portfolio optimization problem as a Quadratic Program (QP) for classical optimization.
5. To convert the optimization problem into a Quadratic Unconstrained Binary Optimization (QUBO) model suitable for quantum optimization.
6. To implement a classical optimization approach using the Trust-Constr optimization algorithm for continuous portfolio weight optimization.
7. To implement a quantum optimization approach using the Quantum Approximate Optimization Algorithm (QAOA) for optimal asset selection.
8. To compare the performance of classical and quantum optimization techniques based on portfolio return, risk, execution time, transaction cost, and investment allocation.
9. To develop an interactive Streamlit dashboard for portfolio visualization, objective function analysis, and optimization comparison.
10. To integrate an AI-powered Portfolio Copilot capable of providing portfolio insights, answering investment-related queries, and generating professional PDF reports.

The overall objective is to create an intelligent portfolio optimization framework that combines financial modeling, optimization algorithms, quantum computing, and artificial intelligence into a unified decision-support system.

Proposed Solution

To address the challenges of multi-asset portfolio construction, this project proposes a **Hybrid Classical-Quantum Multi-Asset Portfolio Optimization Framework** that combines classical optimization techniques with quantum computing to generate efficient portfolio allocations under realistic financial constraints.

Historical market data is collected and preprocessed to compute expected returns, covariance matrices, asset correlations, liquidity scores, and transaction costs. These financial metrics are used to formulate a custom portfolio objective function that incorporates expected return, portfolio risk, transaction cost, liquidity, diversification, and cash utilization.

The project implements two complementary optimization approaches. In the **classical approach**, the portfolio optimization problem is formulated using a **Quadratic Program** based on the custom objective function. The binary asset selection is followed by continuous portfolio weight optimization using the **Trust-Constr** optimization algorithm while satisfying the portfolio constraints.

In the **quantum approach**, the Quadratic Program is converted into a **Quadratic Unconstrained Binary Optimization (QUBO)** model and subsequently transformed into an **Ising Hamiltonian**. The optimization problem is solved using the **Quantum Approximate Optimization Algorithm (QAOA)** to identify the optimal subset of assets. The selected assets are then refined through a classical continuous optimization stage using the **Trust-Constr** optimizer to determine the optimal investment weights.

The optimized portfolios are presented through an interactive **Streamlit dashboard** featuring portfolio allocation, objective function analysis, classical-versus-quantum comparisons, QAOA performance metrics, and an AI-powered Portfolio Copilot capable of generating portfolio insights and downloadable PDF reports.

This hybrid framework integrates financial modeling, classical optimization, quantum optimization, interactive visualization, and artificial intelligence into a unified decision-support system for multi-asset portfolio optimization.

Financial Concepts

Modern Portfolio Theory (Markowitz Portfolio Theory) :

Modern Portfolio Theory (MPT), introduced by Harry Markowitz, provides the foundation for portfolio optimization. The theory aims to maximize expected portfolio returns while minimizing investment risk through diversification. Portfolio risk is measured using the covariance among asset returns, allowing investors to construct efficient portfolios that provide the highest possible return for a given level of risk.

- **Expected Returns:** The anticipated average annual return of an asset based on historical performance. It is the primary measure of profitability used in the optimization objective.
- **Portfolio Risk:** The uncertainty in investment returns, quantified using the covariance matrix of asset returns. It captures both individual asset volatility and inter-asset relationships.
- **Covariance Matrix:** Describes how asset pairs move relative to each other — positive values indicate co-movement, negative values indicate opposite movement. It is central to estimating portfolio risk.
- **Correlation Matrix:** A standardized (-1 to 1) measure of the strength of relationship between asset returns. It helps identify diversified assets that reduce overall risk.
- **Liquidity:** The ease of buying or selling an asset without significantly affecting its price. Higher liquidity is rewarded in the objective function to lower execution costs and enable faster rebalancing.
- **Transaction Cost:** Expenses incurred from trading, including brokerage, taxes, and market impact. These are penalized in the objective function to minimize unnecessary trading.
- **Diversification:** Spreading investments across assets or sectors to reduce risk from individual asset failures. The model rewards portfolios that distribute exposure across sectors.
- **Sector Allocation:** Controls the proportion of capital assigned to industry sectors like Technology, Healthcare, and Energy. Sector constraints prevent concentration risk and maintain balanced exposure.
- **Cash Utilization:** Measures the proportion of capital actively invested versus left idle. The model minimizes unused capital to improve investment efficiency.

Mathematical Formulation

1. Expected Portfolio Return:

$$R_p = \sum_{i=1}^N w_i \mu_i$$

2. Portfolio Risk(Variance):

$$\sigma_p^2 = w^T \Sigma w$$

3. Portfolio Volatility:

$$\sigma_p = \sqrt{w^T \Sigma w}$$

4. Transaction Cost:

$$TC = \sum_{i=1}^N c_i |w_i - w_i^{old}|$$

5. Liquidity:

$$L = \sum_{i=1}^N l_i w_i$$

6. Diversification:

$$D = \sum_{i=1}^N \sum_{j=1}^N (1 - \rho_{ij}) w_i w_j$$

7. Cash Utilization:

$$Cash = 1 - \sum_{i=1}^N w_i$$

8. Objective Function:

$$\text{Obj}_{fn} = R - \lambda \sigma_p^2 - \alpha T + \beta L + \gamma D - \delta C$$

9. Quadratic Program:

$$\min_w \frac{1}{2} w^T Q w + c^T w$$

10.QUBO Formulation:

$$\min_x x^T Q x + c^T x$$

11.Ising Hamiltonian:

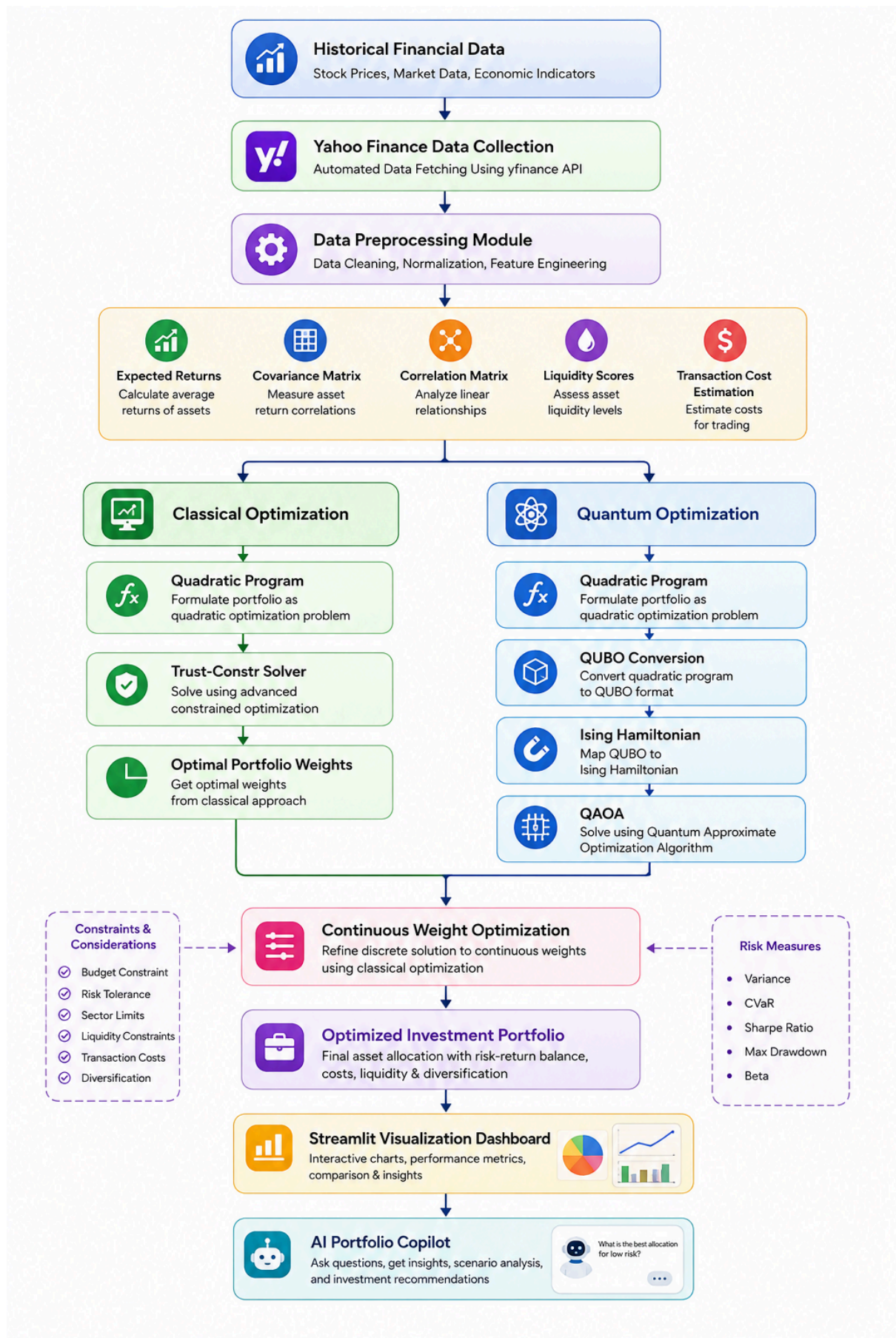
$$H = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j$$

System Architecture

The proposed Hybrid Classical-Quantum Multi-Asset Portfolio Optimization framework consists of multiple interconnected modules that work together to construct optimized investment portfolios. The system integrates financial data processing, classical optimization, quantum optimization, visualization, and artificial intelligence into a unified decision-support platform.

The workflow begins with collecting historical financial data from Yahoo Finance. The collected data is processed to compute essential financial metrics such as expected returns, covariance matrices, correlation matrices, liquidity scores, and transaction cost estimates. These metrics are then used to formulate both the classical optimization model and the quantum optimization model.

The classical optimization module solves the portfolio allocation problem using quadratic programming and continuous optimization techniques. Simultaneously, the quantum optimization module converts the optimization problem into a QUBO formulation and employs QAOA to determine the optimal subset of assets. Finally, the optimized portfolios are visualized through an interactive Streamlit dashboard, while the AI Portfolio Copilot provides portfolio explanations, investment insights, and automated report generation.



Data Collection and Preprocessing

Accurate financial inputs are essential to optimization quality. This module prepares all required data for both optimization models.

Historical Data Collection

Historical adjusted closing prices are collected via the Yahoo Finance API for all selected assets over a user-specified period.

Asset Selection

Assets span multiple sectors: Technology, Healthcare, Financial Services, Energy & Commodities, and ETFs & Index Funds.

Daily Returns

Daily returns are calculated as the percentage change in an asset's price from one trading day to the next, forming the basis for all downstream return and risk calculations.

Annualized Expected Returns

The average daily return for each asset is scaled up to an annual figure using the standard 252 trading days in a year, giving a forward-looking estimate of expected performance.

Covariance Matrix

Measures joint variability of asset returns; serves as the primary risk measure.

Correlation Matrix

Quantifies pairwise relationships between assets; used for diversification analysis.

Liquidity Score

A normalized score based on average trading volume, encouraging the optimizer to favor easily tradable assets.

Transaction Cost Estimation

Estimated using predefined cost coefficients per asset, incorporated as a penalty in the objective function.



FINANCIAL DATA PREPARATION MODULE

This module prepares the financial inputs required for both optimization models by collecting and processing accurate market data.



10.1 Historical Data Collection

Historical adjusted closing prices are collected using the Yahoo Finance API for all selected assets over the user-specified time period.



10.2 Asset Selection

The project supports assets from multiple financial sectors, including:

- Technology
- Healthcare
- Financial Services
- Energy & Commodities
- ETFs & Index Funds

10.3 Daily Returns

Daily returns are computed using:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t is the adjusted closing price on day t .

Date	Price (P_t)	Return (r_t)
Day t-2	100.50	-
Day t-1	101.20	$(101.20 - 100.50) / 100.50 = 0.00697$
Day t	102.10	$(102.10 - 101.20) / 101.20 = 0.00889$
...	...	...

10.4 Annualized Expected Returns

Expected annual returns are calculated as

$$\mu = \bar{r} \times 252$$

where $\bar{r}$ is the average daily return and 252 represents the average number of trading days in one year.



10.5 Covariance Matrix

The covariance matrix measures the joint variability of asset returns and serves as the primary risk measure.

	A	B	C	D	E
A	0.0400	0.0152	0.0101	0.0087	-0.0021
B	0.0152	0.0225	0.0123	0.0065	-0.0010
C	0.0101	0.0123	0.0301	0.0110	-0.0005
D	0.0087	0.0065	0.0110	0.0280	-0.0012
E	-0.0021	-0.0010	-0.0005	-0.0012	0.0180

Low High

10.6 Correlation Matrix

The correlation matrix quantifies the relationship between asset pairs and is used for diversification analysis.

	A	B	C	D	E
A	1.00	0.65	0.32	0.28	-0.15
B	0.65	1.00	0.55	0.20	-0.10
C	0.32	0.55	1.00	0.40	-0.005
D	0.28	0.20	0.40	1.00	-0.12
E	-0.15	-0.10	-0.005	-0.12	1.00

-1 0 1

10.7 Liquidity Score

Each asset is assigned a normalized liquidity score based on its average trading volume, encouraging the optimizer to prefer assets that are easier to trade.

Asset	Avg. Daily Volume	Liquidity Score (0 to 1)
A	12,500,000	1.00
B	6,300,000	0.72
C	2,800,000	0.42
D	1,200,000	0.21
E	500,000	0.08

$$\text{Liquidity Score} = \frac{\text{Volume}_i - \text{Min}(\text{Volume})}{\text{Max}(\text{Volume}) - \text{Min}(\text{Volume})}$$

10.8 Transaction Cost Estimation

Transaction costs are estimated using predefined trading cost coefficients assigned to each asset and incorporated into the objective function as a penalty.

$$TC_i = c_i \times |\Delta w_i|$$

where c_i = cost coefficient
 Δw_i = change in weight

Asset	Cost Coefficient (c_i)	Description
A	0.0010	Low cost (High liquidity)
B	0.0015	Low-Medium cost
C	0.0025	Medium cost
D	0.0035	High cost (Low liquidity)
E	0.0040	Very High cost

These prepared financial inputs are used by both Classical and Quantum Optimization models to construct the objective function with return, risk, liquidity and transaction cost considerations.

Classical Portfolio Optimization

The classical optimization approach formulates the portfolio optimization problem as a **Quadratic Programming (QP)** problem using the proposed multi-objective function. The objective function simultaneously maximizes expected portfolio return while minimizing portfolio risk and transaction costs, rewarding liquidity and diversification, and penalizing unused capital.

Since the optimization problem consists of both **asset selection** and **continuous capital allocation**, the classical framework employs a two-stage optimization process.

Stage 1: Binary Asset Selection

The objective function is first formulated as a **Quadratic Program (QP)** with binary decision variables representing whether an asset is selected in the portfolio. The **CPLEX Optimizer** is used to solve this binary optimization problem and identify the optimal subset of assets while satisfying portfolio constraints.

Stage 2: Continuous Weight Optimization

After the optimal asset subset is obtained, a second optimization stage determines the investment weight assigned to each selected asset. This continuous optimization problem is solved using the **Trust-Constr optimizer**, which computes the optimal portfolio weights while satisfying constraints such as budget allocation, sector allocation limits, and non-negative asset weights.

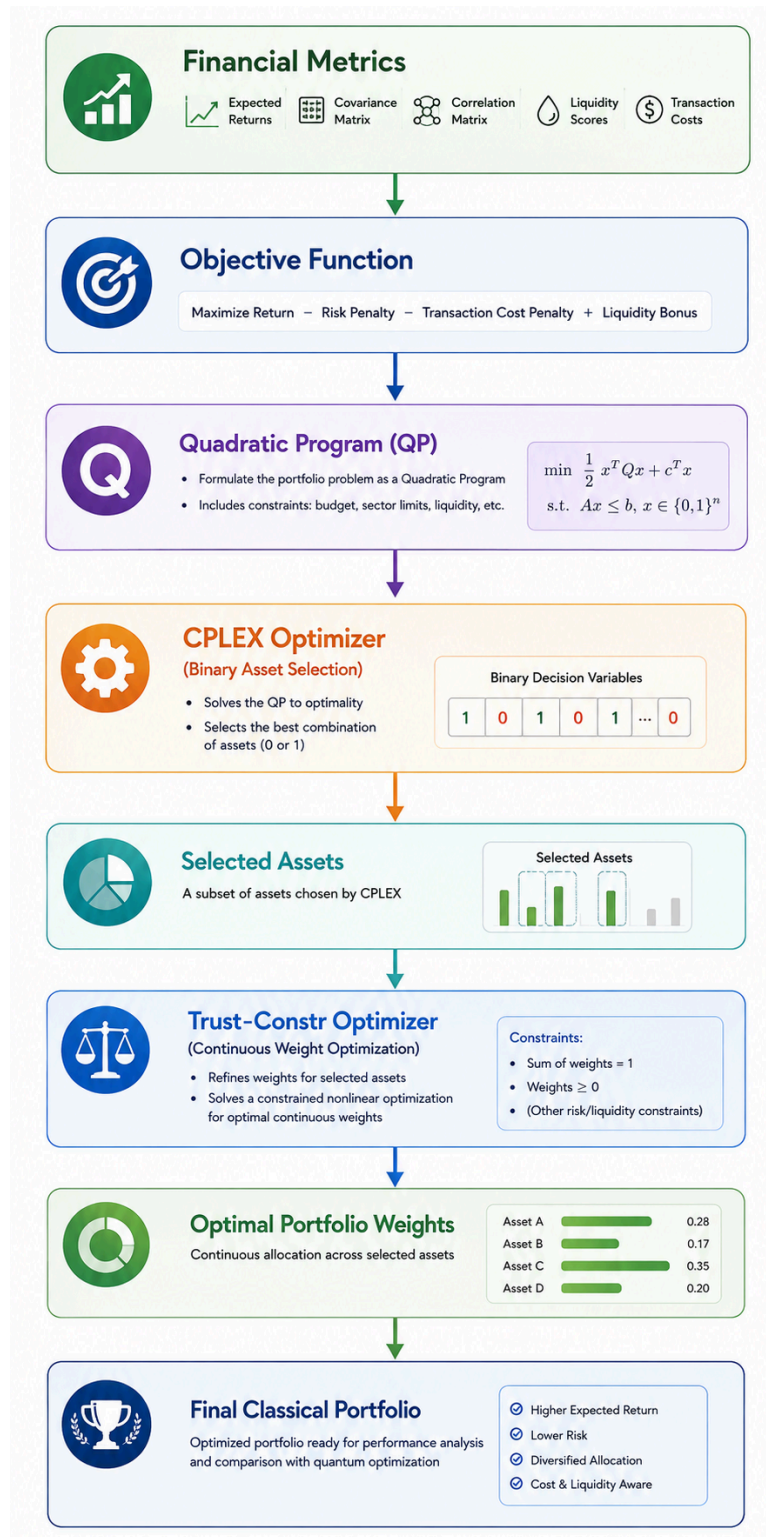
This two-stage optimization strategy separates the asset selection problem from the capital allocation problem, resulting in an efficient and practical portfolio construction framework.

Constraints

The optimizer satisfies the following constraints:

- Budget constraint
- Maximum asset selection
- Sector allocation limits
- Non-negative weights
- Portfolio weight bounds
- Cash utilization constraint

Classical Portfolio Optimization Pipeline



Quantum Portfolio Optimization

The quantum optimization approach formulates the portfolio optimization problem as a Quadratic Programming (QP) problem using the proposed multi-objective function. The objective function incorporates expected portfolio return, portfolio risk, transaction cost, liquidity, diversification, and cash utilization. The Quadratic Program is then transformed into a Quadratic Unconstrained Binary Optimization (QUBO) model to enable execution on a quantum optimization algorithm.

Similar to the classical approach, the quantum framework employs a two-stage optimization process.

Stage 1: Quantum Asset Selection

The **Quadratic Program** is converted into a **QUBO model**, which is subsequently transformed into an Ising Hamiltonian. The **Quantum Approximate Optimization Algorithm (QAOA)** implemented using Qiskit is applied to solve the binary optimization problem. QAOA identifies the optimal subset of assets by minimizing the Ising Hamiltonian while maximizing the portfolio objective function.

Stage 2: Continuous Weight Optimization

The asset subset selected by QAOA is passed to a second optimization stage, where the **SLSQP (Sequential Least Squares Programming)** optimizer determines the optimal continuous investment weights. This stage satisfies portfolio constraints such as budget allocation, sector allocation limits, and portfolio weight bounds while maximizing the overall portfolio performance.

The hybrid optimization strategy combines the quantum search capability of QAOA for asset selection with the efficiency of classical continuous optimization for portfolio weight allocation, producing an optimized and practical investment portfolio.

Quantum Optimization Stages

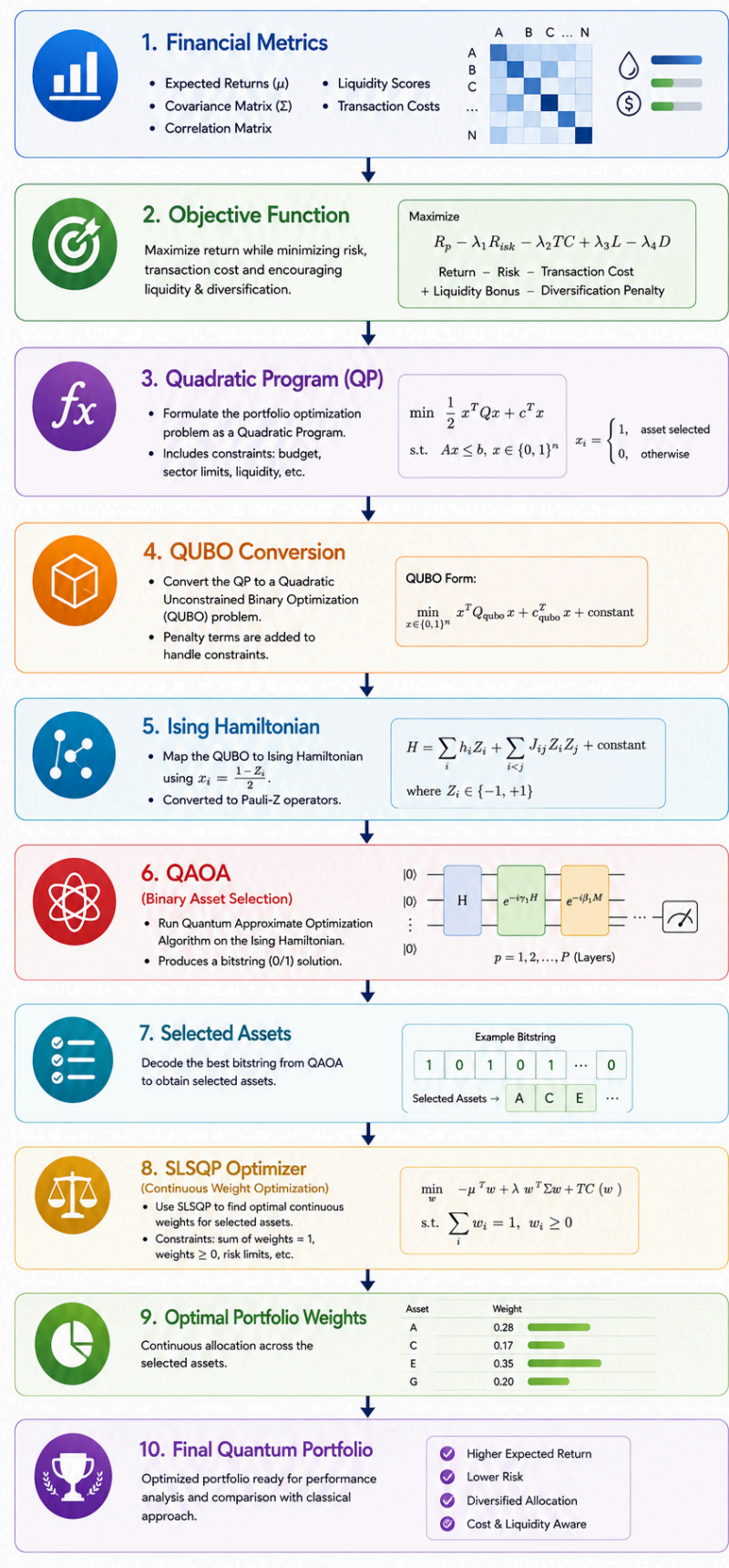
Stage 1: Quantum Optimization

- Construct the Quadratic Program (QP).
- Convert the QP into a QUBO formulation.
- Transform the QUBO into an Ising Hamiltonian.
- Apply QAOA to determine the optimal subset of assets.

Stage 2: Continuous Optimization

- Extract the assets selected by QAOA.
- Optimize the continuous investment weights using the SLSQP optimizer.
- Generate the final portfolio allocation while satisfying investment constraints.

Quantum Portfolio Optimization Pipeline



Dashboard Implementation

Home Page:

Multi Asset Portfolio Optimization

Welcome to the **Multi-Asset Portfolio Optimization Dashboard**, an intelligent investment platform that combines **Modern Portfolio Theory (Markowitz Model)** with **Quantum Optimization techniques** to construct efficient investment portfolios.

Financial markets contain thousands of investment opportunities across stocks, ETFs, commodities, and fixed-income assets. Selecting the optimal combination while balancing **expected returns**, **risk**, **transaction costs**, and **investment constraints** is a complex optimization problem.

This project addresses that challenge using a **hybrid classical-quantum workflow**:

-  Classical Optimization using **Markowitz Portfolio Theory**
-  Quantum Optimization using **QUBO (Quadratic Unconstrained Binary Optimization)**
-  Portfolio solution through **QAOA (Quantum Approximate Optimization Algorithm)**
-  Risk analysis using the covariance matrix
-  Capital allocation and portfolio weight optimization
-  Transaction cost modeling
-  Performance comparison between Classical and Quantum solutions

The dashboard provides interactive visualizations for financial data analysis, portfolio construction, optimization results, and performance evaluation, making it easier to understand how quantum computing can assist in solving real-world financial optimization problems.

Page Description:

Welcome to the Multi-Asset Portfolio Optimization Dashboard.

Imagine you have a fixed amount of money to invest. The challenge is deciding which assets to invest in and how much money to allocate to each one so that you can earn the highest possible return while keeping the risk under control. As the number of assets increases, finding the best combination becomes a complex optimization problem.

This dashboard solves that problem using a hybrid classical and quantum approach. It first analyzes historical market data to calculate expected returns and risk. The portfolio optimization problem is then converted into a QUBO model and solved using the Quantum Approximate Optimization Algorithm (QAOA) to identify the best set of assets. Finally, a classical optimizer (SLSQP) determines the optimal investment weights for the selected assets.

This dashboard includes:

-  Financial data analysis
-  Return and risk calculation
-  Quantum asset selection using QAOA
-  Portfolio weight optimization

-  Transaction cost and liquidity modeling
-  Risk analysis using VaR and CVaR
-  Comparison of Classical and Quantum optimization results

The interactive visualizations help you understand how quantum computing can improve portfolio optimization and support smarter investment decisions.

Portfolio Configuration Page:

The Portfolio Configuration page is where users define all the inputs required for portfolio optimization. It allows users to customize their investment strategy before running the hybrid classical-quantum optimization process.

Investment Capital

The user enters the total amount available for investment. This serves as the budget that will be allocated across the selected assets.

Market Data Period

Users choose the historical start and end dates used to fetch financial data. This data is used to calculate expected returns, volatility, and the covariance matrix.

Portfolio Parameters

These sliders control the optimization objective:

- Risk Aversion (λ): Adjusts the balance between maximizing returns and minimizing risk.
- Liquidity Weight: Gives preference to assets that are easier to buy and sell.
- Transaction Cost: Accounts for trading expenses during portfolio construction.
- Diversification Weight: Encourages investment across different assets instead of concentrating capital in a few holdings.

Asset Selection

Users can select assets from multiple sectors such as:

-  Technology
-  Financial
-  ETFs
-  Healthcare
-  Energy & Commodities

This enables the construction of a diversified multi-asset portfolio.

Sector Allocation

The sliders on the right specify the maximum percentage of capital that can be invested in each sector. These constraints help control sector exposure and improve diversification.

Portfolio Constraints

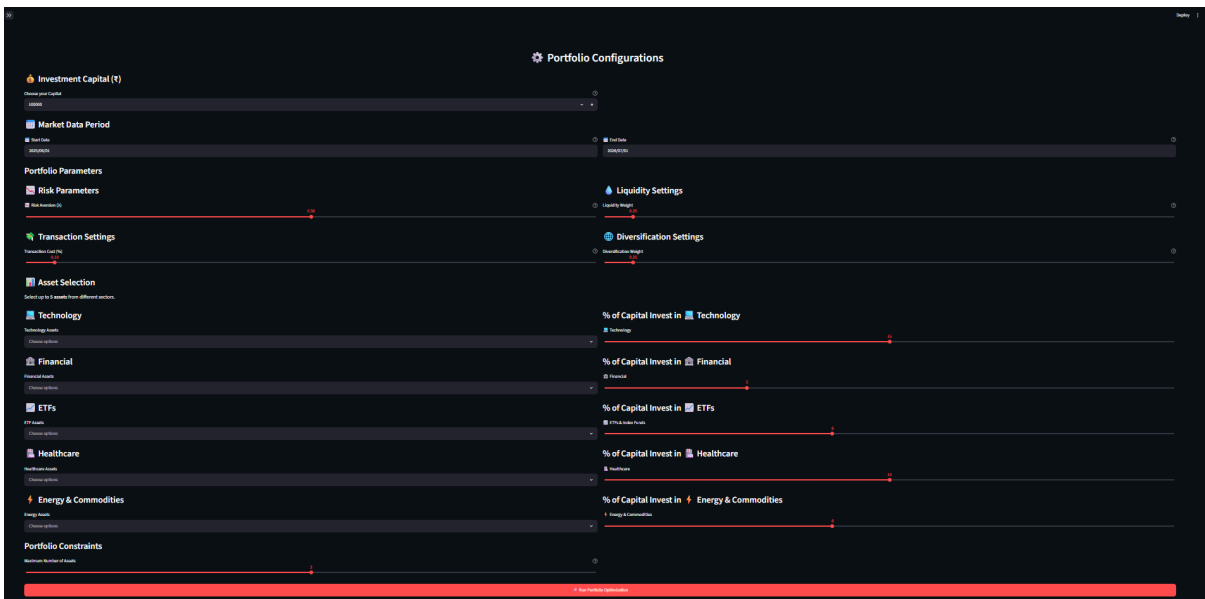
The user specifies the maximum number of assets that can be included in the optimized portfolio. This constraint is incorporated into the optimization model.

Run Portfolio Optimization

Once all parameters and constraints are configured, clicking Run Portfolio Optimization starts the hybrid workflow:

1. Financial data preprocessing
2. Expected return and covariance calculation
3. QUBO formulation
4. Quantum asset selection using QAOA
5. Classical weight optimization using SLSQP
6. Generation of the final optimized portfolio

This page acts as the control center of the application, allowing users to define their investment preferences, risk tolerance, sector limits, and optimization constraints before the quantum-classical optimization engine generates the optimal portfolio.



The screenshot displays the 'Portfolio Configurations' interface, which is a dark-themed control panel for setting investment parameters. The interface is organized into several sections:

- Investment Capital (€):** A dropdown menu set to 'Capital'.
- Market Data Period:** A dropdown menu set to 'Current'.
- Portfolio Parameters:**
 - Risk Parameters:** A slider for 'Risk Tolerance (%)' ranging from 0 to 100, currently set at 50.
 - Liquidity Settings:** A slider for 'Liquidity Weight' ranging from 0 to 100, currently set at 50.
 - Diversification Settings:** A slider for 'Diversification Weight' ranging from 0 to 100, currently set at 50.
- Asset Selection:** A section titled 'Select up to 10 assets from different sectors' with a list of sectors: Technology, Financial, ETFs, Healthcare, and Energy & Commodities. Each sector has a 'Choose options' button.
- Portfolio Constraints:** A slider for 'Maximum Number of Assets' ranging from 0 to 100, currently set at 50.

On the right side of the interface, there are five horizontal sliders representing the percentage of capital investment for each sector:

- % of Capital Invest in Technology:** Slider ranging from 0 to 100, currently set at 50.
- % of Capital Invest in Financial:** Slider ranging from 0 to 100, currently set at 50.
- % of Capital Invest in ETFs:** Slider ranging from 0 to 100, currently set at 50.
- % of Capital Invest in Healthcare:** Slider ranging from 0 to 100, currently set at 50.
- % of Capital Invest in Energy & Commodities:** Slider ranging from 0 to 100, currently set at 50.

At the bottom of the interface, there is a red progress bar and a button labeled 'Run Portfolio Optimization'.

Market Data Analysis Page:

The Market Data Analysis page provides a comprehensive overview of the financial data used for portfolio optimization. It helps users understand the behavior of the selected assets before the optimization process begins.

Expected Asset Returns

This table and bar chart display the expected return of each selected asset, calculated from historical market data. Positive values indicate assets with higher expected growth, while negative values indicate assets with lower expected performance during the selected period.

Asset Covariance Matrix

The covariance heatmap shows how the returns of different assets move together. It is used to measure portfolio risk and forms the foundation of the Markowitz optimization model.

Asset Correlation Matrix

The correlation matrix illustrates the strength of the relationship between asset returns. Highly correlated assets move in similar directions, while low or negatively correlated assets provide better diversification opportunities.

Historical Asset Prices

This line chart displays the historical price movement of each selected asset over the chosen time period, allowing users to compare trends, volatility, and overall market performance.

Efficient Frontier Analysis

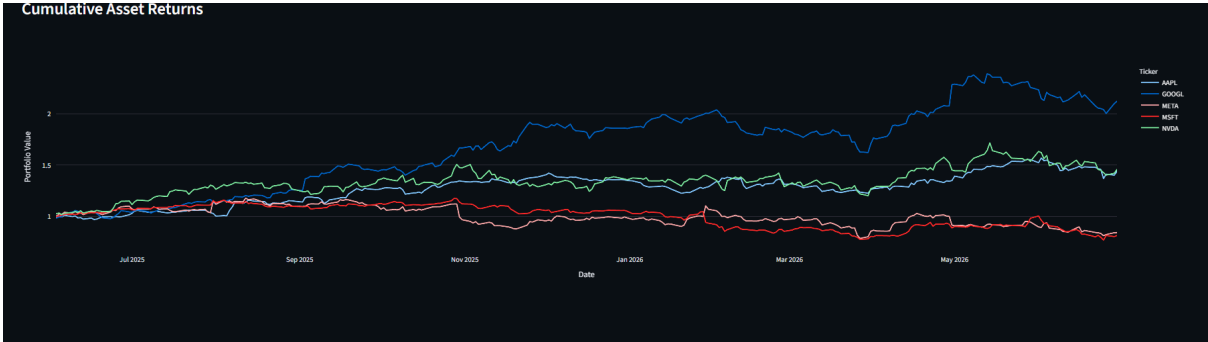
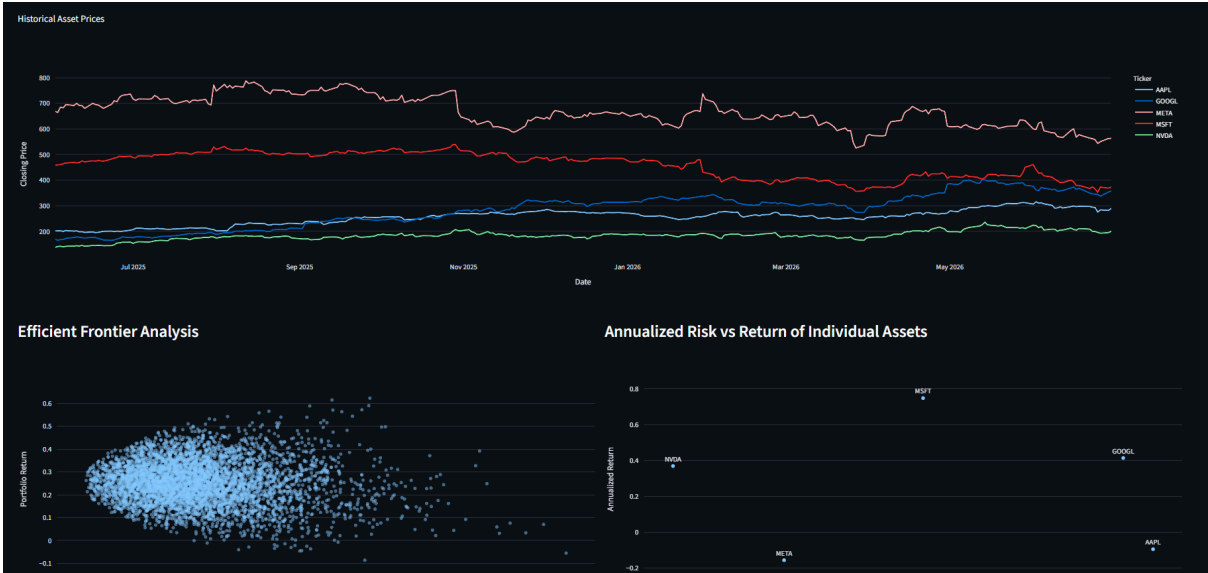
The Efficient Frontier visualizes thousands of possible portfolios, showing the trade-off between expected return and portfolio risk. It helps identify portfolios that offer the highest return for a given level of risk.

Annualized Risk vs Return

This scatter plot compares individual assets based on their annualized return and annualized risk (volatility). It enables users to evaluate the performance of each asset and understand the balance between potential return and investment risk.

Overall, this page provides the financial insights required to evaluate market behavior and serves as the foundation for both the classical and quantum portfolio optimization process.

Market Data Analysis Dashboard



Classical Portfolio Optimization Page:

The Classical Portfolio Optimization page presents the results obtained using a traditional optimization approach. It combines a binary optimizer (CPLEX) to select the optimal assets and a continuous optimizer (TRUST-CONSTR) to determine the optimal investment weights for the selected portfolio. The page also provides detailed performance metrics and visualizations to help evaluate the portfolio.



Portfolio Summary

This section displays the key performance metrics of the optimized portfolio, including:

- Expected Return – The estimated portfolio return.
- Expected Risk – The portfolio risk calculated from the covariance matrix.
- Investment Capital – The total capital available for investment.
- Selected Assets – Assets chosen by the binary optimizer.
- Binary Optimizer – CPLEX, used for asset selection.
- Continuous Optimizer – TRUST-CONSTR, used for optimizing portfolio weights.
- Invested Capital & Remaining Cash – Amount invested and unallocated capital.
- Value at Risk (VaR) and Conditional Value at Risk (CVaR) – Downside risk measures.
- Expected Profit – Estimated profit based on the optimized portfolio.



Objective Function Breakdown

The dashboard displays the contribution of each component in the optimization objective, such as:

- Expected return
- Risk penalty
- Transaction cost
- Liquidity reward
- Diversification penalty
- Overall optimization objective

This helps users understand how each factor influences the final portfolio.



Portfolio Allocation

These charts illustrate how the investment capital is distributed:

- Portfolio Weight Distribution shows the percentage allocated to each selected asset.
- Investment Allocation displays the actual capital invested in each asset.
- Expected Profit per Asset highlights the contribution of each asset to the portfolio's expected profit.

Risk Analysis

The page includes several visualizations for evaluating portfolio risk:

- Covariance Heatmap showing how asset returns vary together.
- Correlation Heatmap illustrating the relationship between asset returns.
- Risk Contribution by Asset identifying which assets contribute the most to overall portfolio risk.
- Risk vs Return scatter plot comparing the annualized risk and return of individual assets.

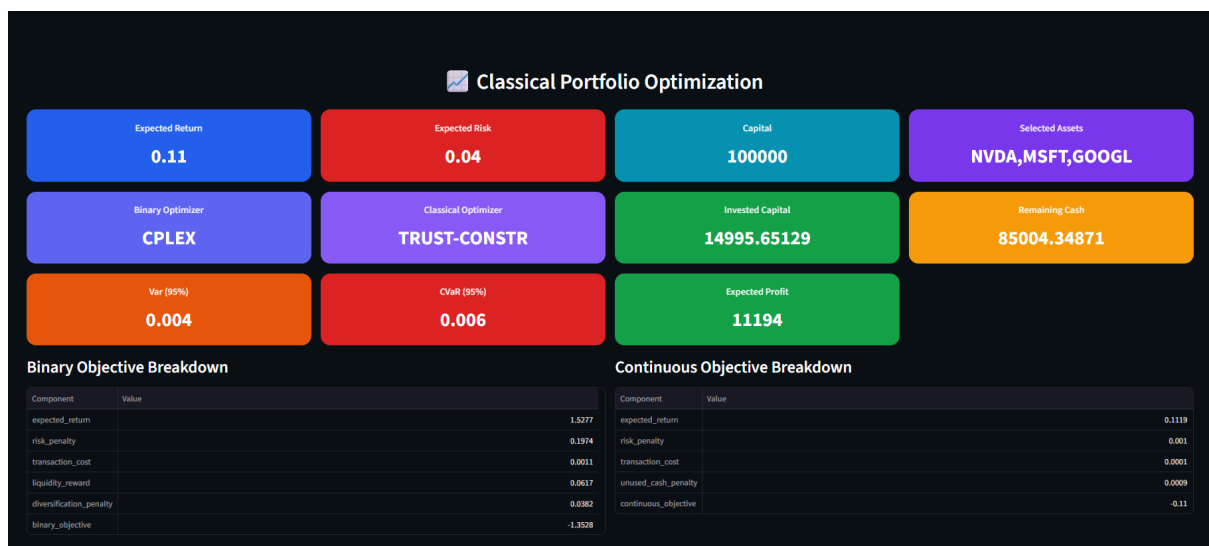
Performance Analysis

Additional performance metrics help assess portfolio quality:

- Sharpe Ratio measures the portfolio's risk-adjusted return.
- Cumulative Portfolio Growth tracks the portfolio value over time based on historical data.
- Portfolio Allocation by Expected Profit visualizes which assets contribute most to the portfolio's projected returns.

Overall, this page provides a complete evaluation of the classically optimized portfolio, allowing users to analyze asset selection, capital allocation, portfolio risk, expected returns, and overall performance before comparing these results with the quantum optimization approach.

Classical Portfolio Optimization Dashboard



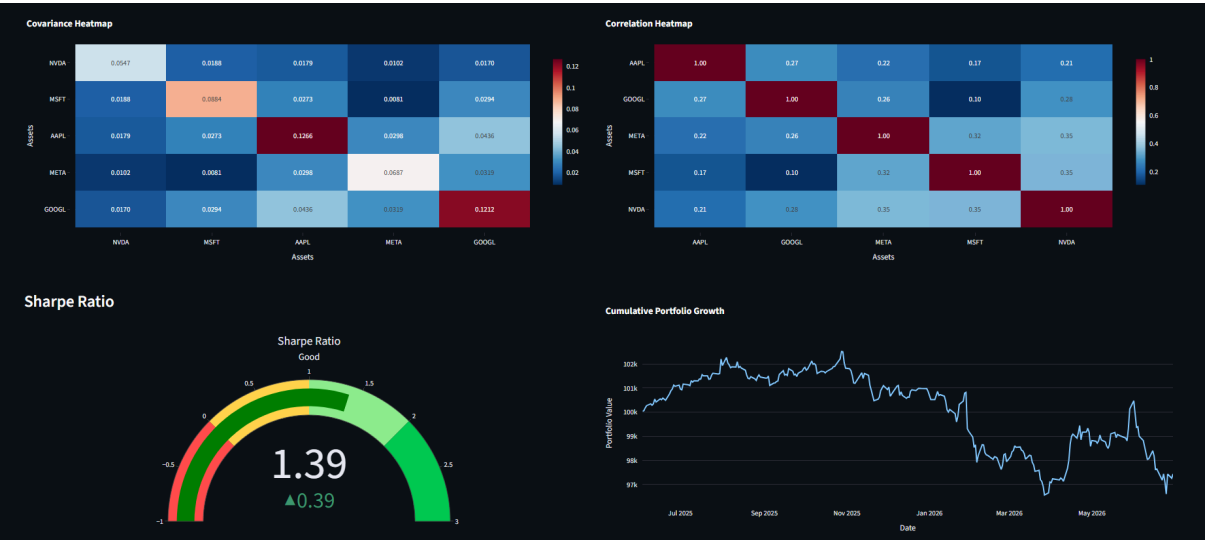


Portfolio Weight Distribution

Asset	Weight (%)
NVDA	0.0%
MSFT	15.0%
AAPL	0.0%
META	0.0%
GOOGL	0.0%

Expected Profit Per Asset

Asset	Expected Profit (€)
NVDA	€3.18
MSFT	€11,187.22
AAPL	€0.00
META	€0.00
GOOGL	€4.05



Sharpe Ratio

Sharpe Ratio: 1.39 (Good)

Cumulative Portfolio Growth



Portfolio Allocation by Expected Profit

Asset	Expected Profit (€)	Allocation (%)
MSFT	14,977.19496	100%
NVDA	3.18	0.02%
AAPL	0.00	0.00%
META	0.00	0.00%
GOOGL	4.05	0.03%

Quantum Portfolio Optimization

The Quantum Optimization Results page presents the optimized portfolio generated using a hybrid quantum-classical workflow. The portfolio optimization problem is first formulated as a QUBO (Quadratic Unconstrained Binary Optimization) model and then solved using the Quantum Approximate Optimization Algorithm (QAOA). The selected assets are further refined using a classical optimizer (COBYLA) to determine the optimal solution. This page provides comprehensive performance metrics, risk analysis, portfolio allocation, and quantum-specific insights to evaluate the effectiveness of the quantum optimization process.

Portfolio Summary

This section displays the key performance metrics of the optimized quantum portfolio, including:

- Portfolio Return – Expected return of the optimized portfolio.
- Portfolio Risk – Overall portfolio volatility.
- Expected Profit – Estimated profit from the optimized portfolio.
- Expected Risk – Risk associated with the selected portfolio.
- Algorithm – QAOA used for quantum optimization.
- Optimizer – COBYLA used to optimize the QAOA parameters.
- Layers – Number of QAOA layers used during optimization.
- Circuit Depth – Depth of the quantum circuit.
- Capital Invested & Remaining Cash – Amount invested and unallocated capital.
- Transaction Cost – Estimated trading cost.
- Objective Value – Final objective value obtained after optimization.
- Value at Risk (VaR) and Conditional Value at Risk (CVaR) – Measures of downside portfolio risk.
- Execution Time – Time required to complete the optimization.
- Selected Assets – Assets chosen by the quantum algorithm.

Objective Function Breakdown

This section explains how the optimization objective is constructed by displaying the contribution of each component:

- Expected Return
- Risk Penalty
- Transaction Cost
- Liquidity Reward
- Diversification Penalty
- Overall Objective Value

This breakdown helps users understand how QAOA balances return, risk, liquidity, diversification, and transaction costs while searching for the optimal portfolio.

Portfolio Allocation

These visualizations show how the optimized investment is distributed across the selected assets:

- Portfolio Weight Distribution displays the percentage allocation of each selected asset.
- Capital Allocation Across Assets shows the investment amount allocated to every asset.
- Expected Profit per Asset illustrates each asset's contribution to the portfolio's expected profit.

Portfolio Performance

This section evaluates the historical performance of the optimized portfolio using:

- Portfolio Growth Over Time to track changes in portfolio value.
- Portfolio Drawdown to measure declines from previous portfolio peaks.
- Rolling Portfolio Volatility to analyze changes in portfolio risk over time.

Risk Analysis

Several visualizations are included to evaluate portfolio risk:

- Covariance Matrix showing how asset returns vary together.
- Correlation Matrix illustrating the relationship between asset returns.
- Efficient Frontier demonstrating the trade-off between portfolio risk and expected return.
- Risk Contribution by Asset identifying how much each asset contributes to total portfolio risk.
- Risk vs Return of Individual Assets comparing the expected return and volatility of each asset.

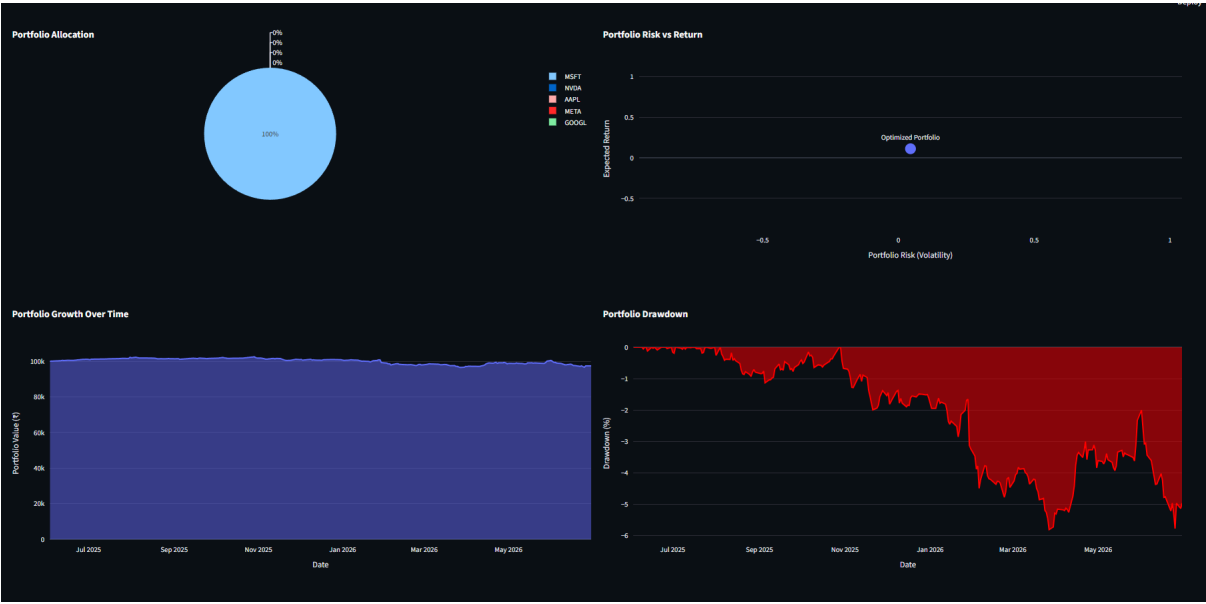
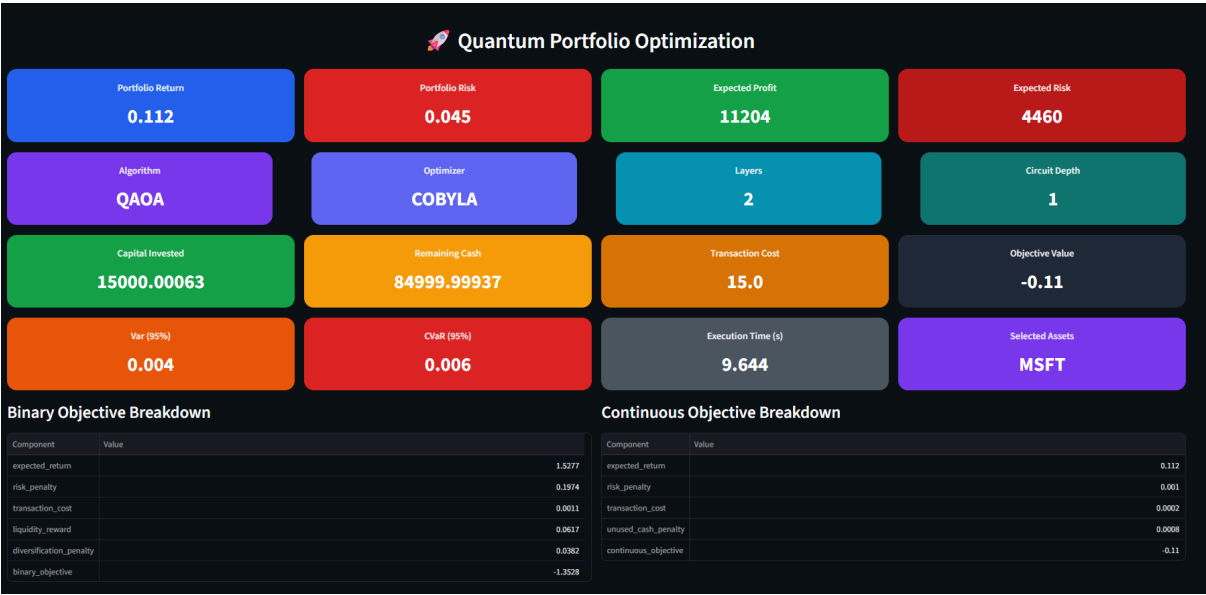
Quantum Analysis

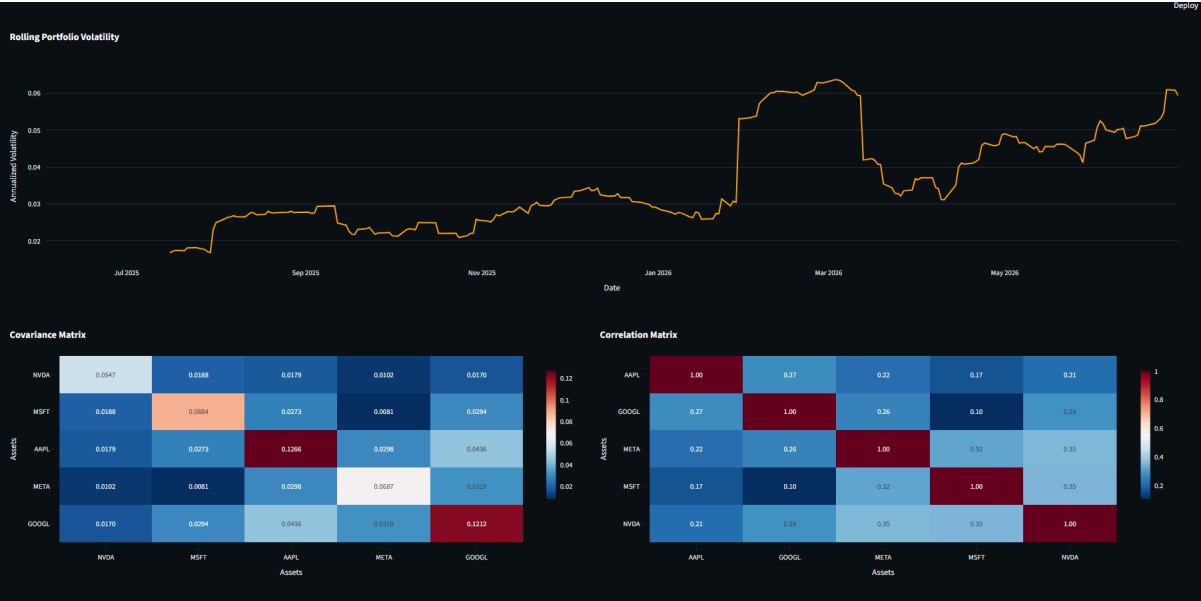
This section provides insights into the quantum optimization process:

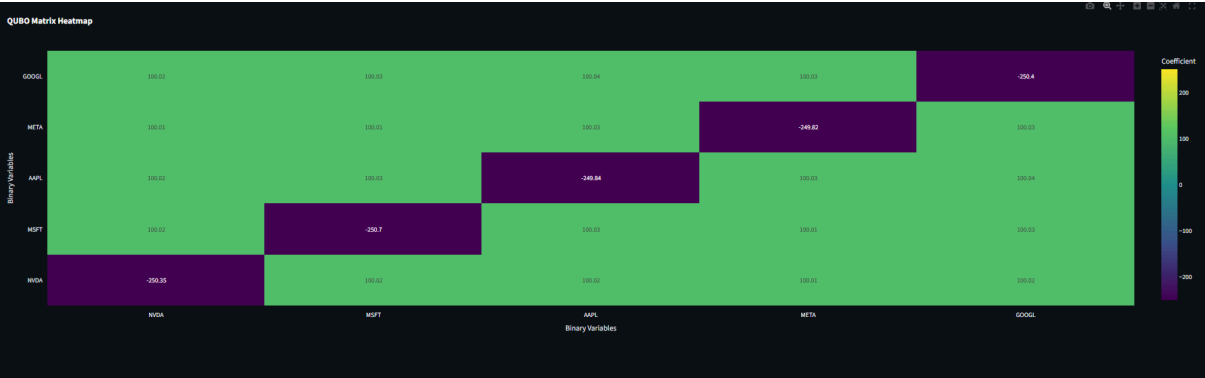
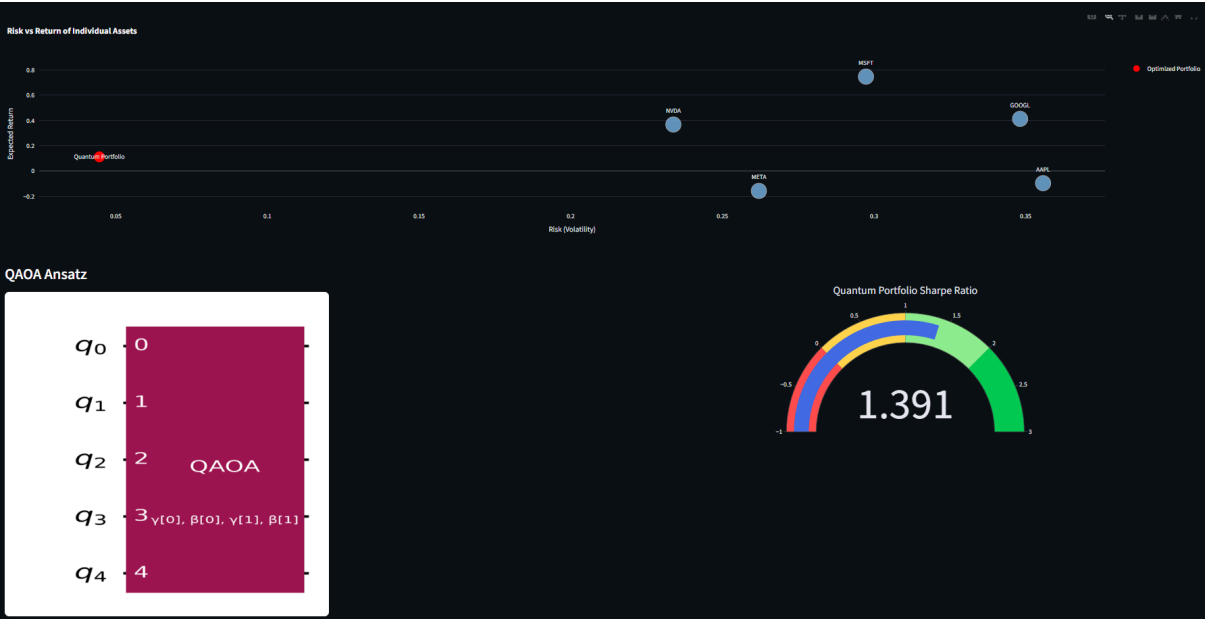
- QAOA Ansatz visualizes the parameterized quantum circuit used during optimization.
- QUBO Matrix Heatmap illustrates the binary optimization problem that is solved by the quantum algorithm before execution on the quantum simulator.

Overall, this page provides a complete evaluation of the Quantum Optimization Results, allowing users to analyze portfolio performance, capital allocation, optimization metrics, portfolio risk, and the behavior of the QAOA algorithm. It also demonstrates how quantum computing can be applied to solve complex portfolio optimization problems and enables direct comparison with the classical optimization approach.

Quantum Portfolio Optimization Dashboard







Classical Vs Quantum Portfolio Comparison

The **Classical vs Quantum Comparison** page provides a side-by-side evaluation of the portfolios generated by the **classical optimization** and **quantum optimization** approaches. It compares their performance, risk, portfolio allocation, and optimization metrics, allowing users to understand the strengths and differences of each method. This comparison helps determine whether the quantum approach offers improvements over the classical solution for portfolio optimization.

Portfolio Summary

This section presents the key performance metrics for both optimization approaches, including:

- **Portfolio Return** – Expected return generated by each portfolio.
- **Portfolio Risk** – Overall portfolio volatility.
- **Algorithm Used** – CPLEX for classical optimization and QAOA for quantum optimization.
- **Classical Optimizer** – TRUST-CONSTR for the classical approach and SLSQP for the quantum workflow.
- **Sharpe Ratio** – Risk-adjusted performance of each portfolio.
- **Expected Profit** – Estimated profit from each optimized portfolio.
- **Portfolio Allocation** – Distribution of capital among the selected assets.

Performance Comparison

This section compares the overall performance of the two portfolios using several visualizations:

- **Risk vs Return Comparison** illustrates the trade-off between expected return and portfolio risk.
- **Portfolio Allocation Comparison** compares how investment capital is allocated across the selected assets.
- **Expected Portfolio Return Comparison** shows the difference in expected returns.
- **Portfolio Risk Comparison** compares the overall risk of each portfolio.

Risk Analysis Comparison

The dashboard compares important downside risk metrics for both portfolios:

- **Value at Risk (VaR)** compares the maximum expected loss at a 95% confidence level.
- **Conditional Value at Risk (CVaR)** compares the average loss beyond the VaR threshold, providing a deeper measure of downside risk.

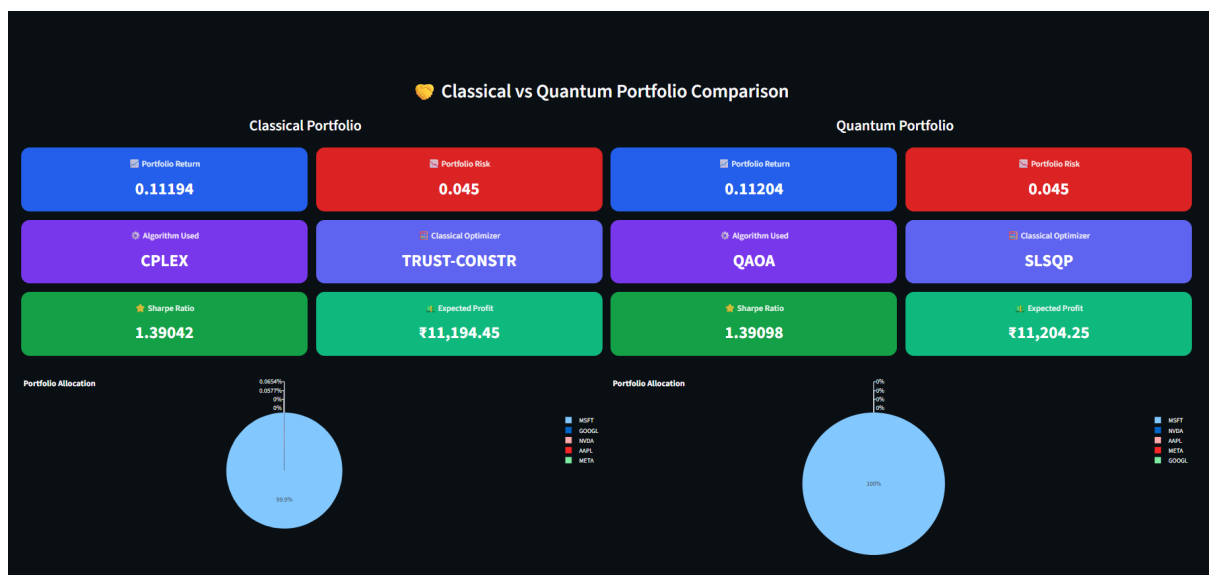
★ Performance Metrics Comparison

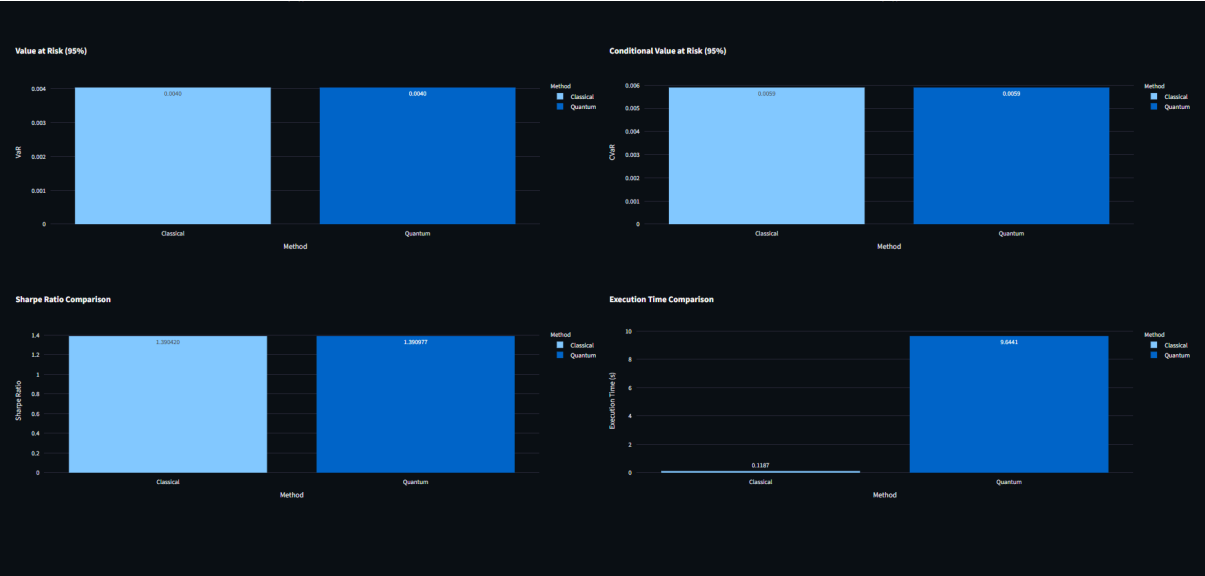
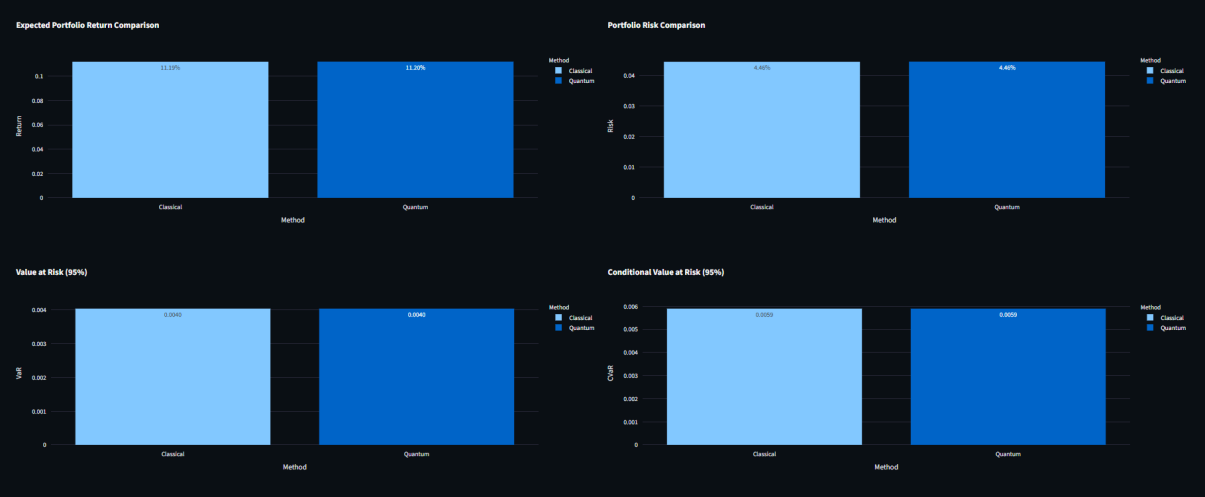
This section evaluates the efficiency and computational performance of both optimization methods through:

- **Sharpe Ratio Comparison** to identify which portfolio delivers better risk-adjusted returns.
- **Execution Time Comparison** to compare the computational time required by the classical and quantum optimization approaches.

Overall, this page provides a comprehensive comparison between **Classical** and **Quantum Portfolio Optimization**, enabling users to evaluate differences in portfolio return, risk, capital allocation, profitability, risk-adjusted performance, and execution time. It helps determine how quantum optimization performs relative to traditional optimization techniques and highlights the potential advantages of hybrid quantum-classical portfolio optimization.

Classical Vs Quantum Dashboard





Portfolio Playground Page:

The Portfolio Objective Playground is an interactive page that allows users to experiment with the portfolio optimization objective function by adjusting key financial parameters and observing their impact in real time. It provides an intuitive way to understand how different optimization weights influence the final objective value and the quality of the optimized portfolio.

Portfolio Parameters

This section allows users to modify the financial characteristics of the portfolio through interactive sliders:

- Expected Return – Represents the anticipated return of the portfolio.
- Portfolio Risk – Measures the overall portfolio volatility.
- Transaction Cost – Represents the cost incurred while trading assets.
- Liquidity Score – Indicates how easily assets can be bought or sold.
- Diversification Score – Measures how well the portfolio is diversified across different assets or sectors.

Optimization Weights

Users can adjust the importance of each objective function component by changing the optimization weights:

- Risk Aversion (λ) – Controls the trade-off between maximizing returns and minimizing risk.
- Transaction Weight (α) – Determines the influence of transaction costs on the optimization.
- Liquidity Weight (β) – Controls the importance of selecting highly liquid assets.
- Diversification Weight (γ) – Determines how strongly diversification is encouraged during optimization.

Objective Function

The dashboard displays the portfolio optimization objective function:

$$f(x) = R - \lambda\sigma^2 - \alpha T + \beta L + \gamma D$$

where:

- R = Expected Return
- σ^2 = Portfolio Risk (Variance)
- T = Transaction Cost
- L = Liquidity Score
- D = Diversification Score

As users modify the parameters or optimization weights, the objective value is recalculated instantly, providing immediate feedback on the optimization process.

 Objective Evaluation

This section displays the current optimization metrics, including:

- Return
- Risk
- Transaction Cost
- Liquidity Score
- Objective Value
- Optimization Quality

These metrics help users understand how each parameter contributes to the final optimization result and whether the portfolio configuration is producing a strong or moderate objective value.

Overall, the Portfolio Objective Playground serves as an interactive learning and experimentation tool, enabling users to explore the relationship between return, risk, transaction costs, liquidity, and diversification. It provides valuable insights into how changes in optimization parameters affect the objective function before running the complete classical or quantum portfolio optimization workflow.

Portfolio Playground Dashboard:



AI - Copilot Page:

The AI Copilot is an intelligent assistant designed to help users understand portfolio optimization concepts, interpret optimization results, and make informed investment decisions. It provides real-time explanations, personalized insights, and guidance on both classical and quantum portfolio optimization, making the platform more interactive and user-friendly.



Interactive Assistant

Users can ask questions related to:

- Portfolio optimization concepts
- Classical and quantum optimization techniques
- Risk and return analysis
- Investment strategies
- Performance evaluation
- Financial metrics such as VaR, CVaR, Sharpe Ratio, and portfolio risk

The AI Copilot generates contextual responses based on the portfolio optimization workflow and financial analysis.



Quick Questions

To simplify user interaction, the dashboard includes predefined questions such as:

- Compare Classical vs Quantum – Explains the differences between classical and quantum optimization results.
- Explain QAOA – Describes how the Quantum Approximate Optimization Algorithm is used for portfolio optimization.
- Analyze Portfolio – Provides insights into portfolio performance, risk, and asset allocation.
- Improve Portfolio – Suggests recommendations for enhancing portfolio performance by adjusting optimization parameters and investment strategies.



Portfolio Summary Report

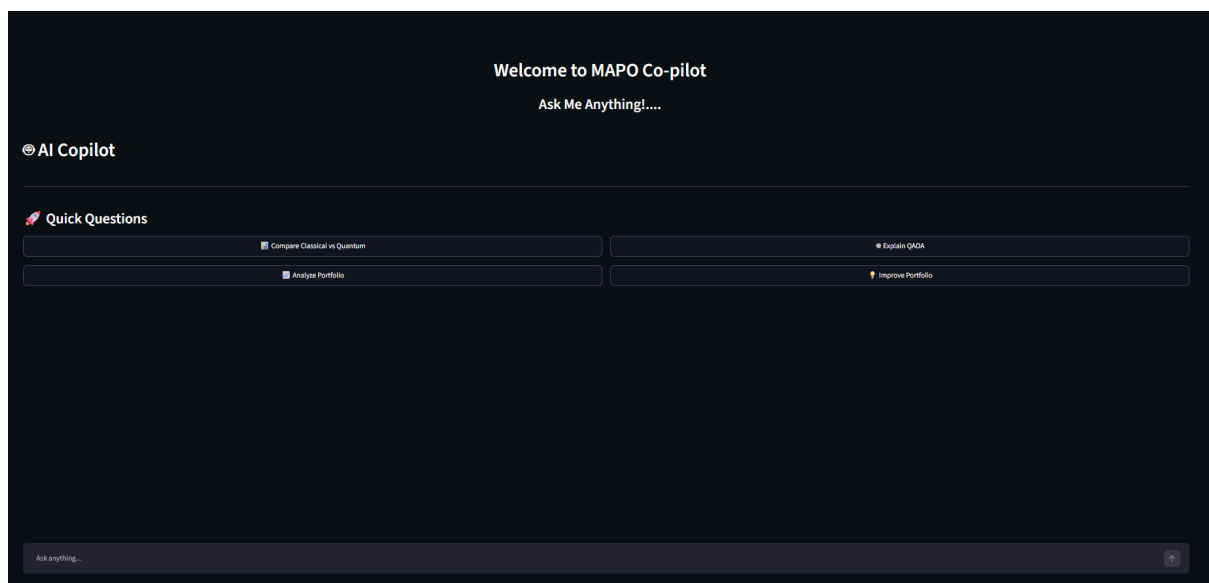
The AI Copilot also provides an option to download a comprehensive PDF summary of the optimized portfolio. The report includes key portfolio metrics, selected assets, capital allocation, expected returns, risk measures, optimization results, and performance analysis. This feature enables users to save, share, or present their portfolio optimization results in a professional format.

Conversational Interface

Users can enter custom questions in the chat box to receive detailed explanations, recommendations, and technical insights. The AI Copilot acts as an interactive guide throughout the portfolio optimization process, helping users better understand the generated results and the underlying optimization techniques.

Overall, the AI Copilot enhances the user experience by combining financial analysis, portfolio optimization, and quantum computing concepts into a conversational assistant while also providing downloadable portfolio reports for documentation and future reference.

AI Copilot Dashboard:



Advantages

The proposed Hybrid Classical-Quantum Multi-Asset Portfolio Optimization framework offers several advantages over conventional portfolio optimization methods by integrating financial modeling, quantum computing, interactive visualization, and artificial intelligence into a single platform.

Hybrid Optimization Framework

The project combines classical optimization techniques with quantum computing to leverage the strengths of both approaches. Classical optimization provides reliable continuous portfolio weight allocation, while quantum optimization efficiently addresses the combinatorial asset selection problem.

Real-World Financial Constraints

Unlike traditional portfolio optimization models that focus only on return and risk, the proposed framework incorporates practical investment considerations, including transaction costs, liquidity, diversification, sector allocation limits, and cash utilization. This enables the generation of realistic and implementable investment portfolios.

Interactive Dashboard

An interactive Streamlit dashboard has been developed to provide a user-friendly interface for portfolio configuration and analysis. Users can visualize portfolio allocations, financial metrics, optimization results, objective function components, and compare classical and quantum solutions through interactive charts and tables.

AI-Assisted Portfolio Analysis

The integrated AI Portfolio Copilot enhances decision-making by interpreting optimization results, answering portfolio-related questions, providing investment insights, and generating comprehensive PDF reports. This improves the explainability and accessibility of the optimization framework.

Explainable Optimization

The project provides detailed objective function breakdowns and optimization metrics, allowing users to understand how individual financial factors such as risk, liquidity, transaction costs, and diversification contribute to the final portfolio allocation.

Modular Architecture

The implementation follows a modular software architecture, separating data processing, optimization algorithms, visualization, and AI components. This design improves code maintainability, scalability, and simplifies the integration of future enhancements.

Limitations

Although the proposed framework successfully demonstrates the application of hybrid classical-quantum optimization for portfolio construction, several limitations remain.

Quantum Simulation

The Quantum Approximate Optimization Algorithm (QAOA) is currently executed using a quantum simulator rather than real quantum hardware. Consequently, the reported execution times and optimization performance may differ from those obtained on physical quantum processors.

Limited Asset Universe

The project has been evaluated using a relatively small set of financial assets due to the current limitations of quantum algorithms and available computational resources. Larger portfolios may require additional optimization techniques and more advanced quantum hardware.

Parameter Sensitivity

The optimization performance depends on several user-defined parameters, including risk aversion, transaction cost weights, liquidity weights, diversification weights, and penalty coefficients. Selecting appropriate parameter values requires careful tuning based on investment objectives.

Historical Data Dependency

Expected returns, covariance matrices, and other financial metrics are estimated using historical market data. Since historical performance does not guarantee future results, portfolio recommendations may not always reflect future market behavior.

Scalability of Current Quantum Hardware

Current quantum computers have limited qubit counts and are susceptible to noise and decoherence. As a result, executing large-scale portfolio optimization problems directly on existing quantum hardware remains challenging.

Conclusion

This project presented a Hybrid Classical-Quantum Multi-Asset Portfolio Optimization framework that combines classical optimization techniques with quantum computing to address real-world portfolio construction problems. The framework integrates financial modeling, optimization algorithms, interactive visualization, and artificial intelligence into a unified decision-support platform.

The optimization process begins with the collection and preprocessing of historical financial market data to compute expected returns, covariance matrices, asset correlations, liquidity scores, and transaction cost estimates. These financial metrics are incorporated into a custom portfolio objective function designed to maximize expected returns while minimizing portfolio risk and transaction costs, rewarding liquidity and diversification, and penalizing unused capital.

The classical optimization approach formulates the problem as a Quadratic Program (QP). Binary asset selection is performed using the CPLEX optimizer, followed by continuous portfolio weight optimization using the Trust-Constr optimization algorithm. This two-stage optimization strategy efficiently determines both the optimal asset subset and the corresponding investment weights.

The quantum optimization approach follows a similar formulation by converting the Quadratic Program into a Quadratic Unconstrained Binary Optimization (QUBO) model and subsequently transforming it into an Ising Hamiltonian. The Quantum Approximate Optimization Algorithm (QAOA) is then employed to perform binary asset selection. The selected assets are further refined using the Sequential Least Squares Programming (SLSQP) optimizer to determine the optimal continuous investment weights, resulting in a hybrid quantum-classical optimization workflow.

To improve usability and interpretability, an interactive Streamlit dashboard was developed, enabling users to configure optimization parameters, visualize portfolio allocations, analyze objective function components, compare classical and quantum optimization results, and monitor optimization performance. The integrated AI Portfolio Copilot further enhances the platform by providing portfolio analysis, answering investment-related questions, and generating comprehensive PDF reports.

Overall, the project demonstrates the practical application of hybrid quantum-classical optimization for multi-asset portfolio construction. It provides a scalable and extensible framework that can evolve alongside advancements in quantum computing, offering a promising direction for future financial optimization and investment decision-support systems.

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[IBM CPLEX Optimizer Documentation](#)

[Yahoo Finance API \(yfinance\) Documentation](#)

[Plotly Python Documentation](#)

[Streamlit Documentation](#)

[NumPy Documentation](#)

[Pandas Documentation](#)

[Matplotlib Documentation](#)

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Vanguard – WISER Quantum Challenge 2026. *Challenge Statement: Multi-Asset Portfolio Construction*. (Project challenge document provided as part of the WISER Quantum Challenge.)

Thank You

Thank you for taking the time to review this project.

Multi-Asset Portfolio Optimization using Hybrid Classical and Quantum Computing demonstrates the integration of financial modeling, optimization techniques, quantum computing, and artificial intelligence into a unified decision-support platform.

This project was developed as part of the WISER Vanguard Quantum Challenge 2026, with the objective of exploring the practical application of hybrid quantum-classical optimization for real-world portfolio management.

We sincerely appreciate your valuable time, feedback, and interest in this work.

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